Question-1:

Two single precision floating point numbers a and b are stored in memory consecutively. The memory location of the first number, a is directly stored in register x19.
Write necessary code that adds these two numbers and stores the result in a variable called d. Variable d is mapped to register f20.

Question-2:

Two single-precision numbers a and b are stored consecutively in memory. The memory location of the first number, a is directly stored in register x19.
Convert the given high level code to corresponding risc-v code:
$res = (a - b)^2$
Note: res is mapped to f16

Question-3:

Convert the given high level code to corresponding risc-v code:

```
if (a > b):
   a = a * d
elif (a == b):
   b -= e;
else:
    arr1[25] = b / e
```

Note: a, b, d, e variables are mapped to f4, f5, f6, f7 registers and all the numbers stored are single precision numbers. arr1 is a word array. The base address of arr1 is in register x20.

Question-4:

Convert the given high level code to corresponding risc-v code:

```
if (a >= b):
    a = (a * c) + d;
else:
   b = b / c;
    arr1[25] = b ;
```

Note: a, b, c, d variables are mapped to f4, f5, f6, f7 registers and all the numbers stored are single precision numbers. arr1 is a word array. The base address of arr1 is in register x20.

You can not use fadd or fmul instructions here.